

TriageSim: A Conversational Emergency Triage Simulation Framework from Structured Electronic Health Records

Dipankar Srirag, Quoc Dung Nguyen, Aditya Joshi, Padmanesan Narasimhan, Salil Kanhere
University of New South Wales, Sydney
d.srirag@unsw.edu.au



Triage is not just a row in a table. It is a 2 minute dialogue.

Research in emergency triage is restricted to **structured electronic health records (EHR)** due to **regulatory constraints** on nurse-patient interactions.

TriageSim generates synthetic, **persona-conditioned triage conversations** from EHRs.

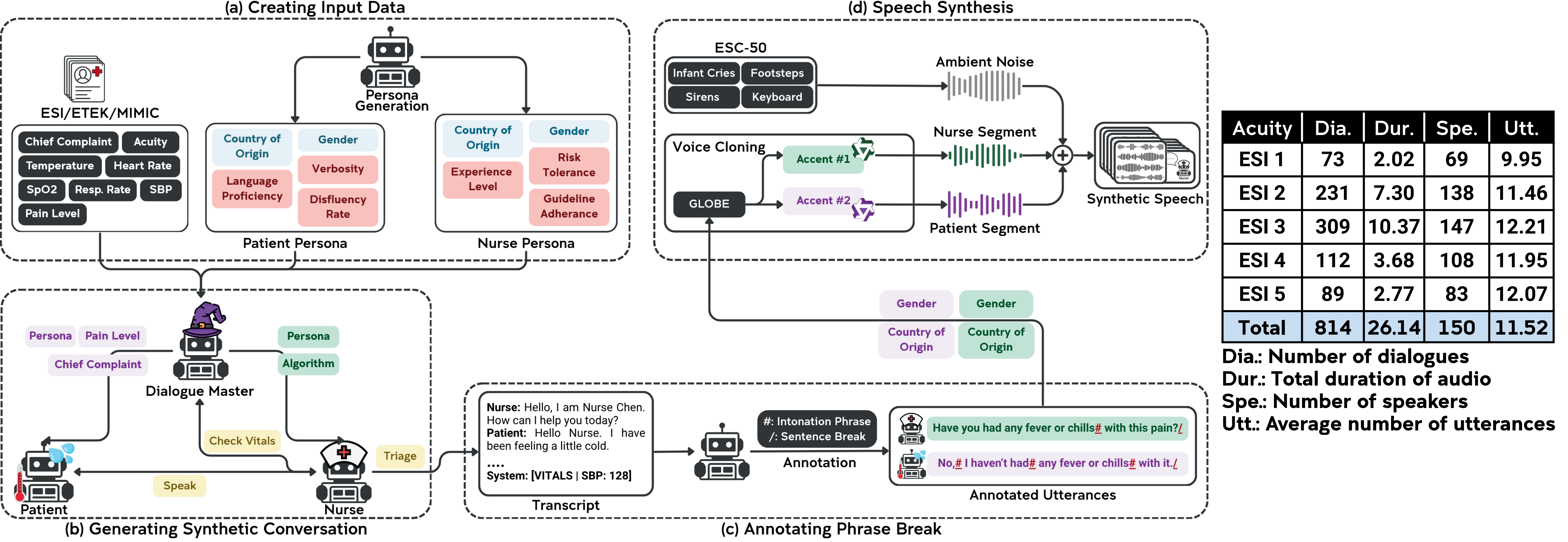
We simulate **814 dialogues** and **26 hours** of audio, with **150 speakers**.

Aditya Joshi
@aadi_joshi

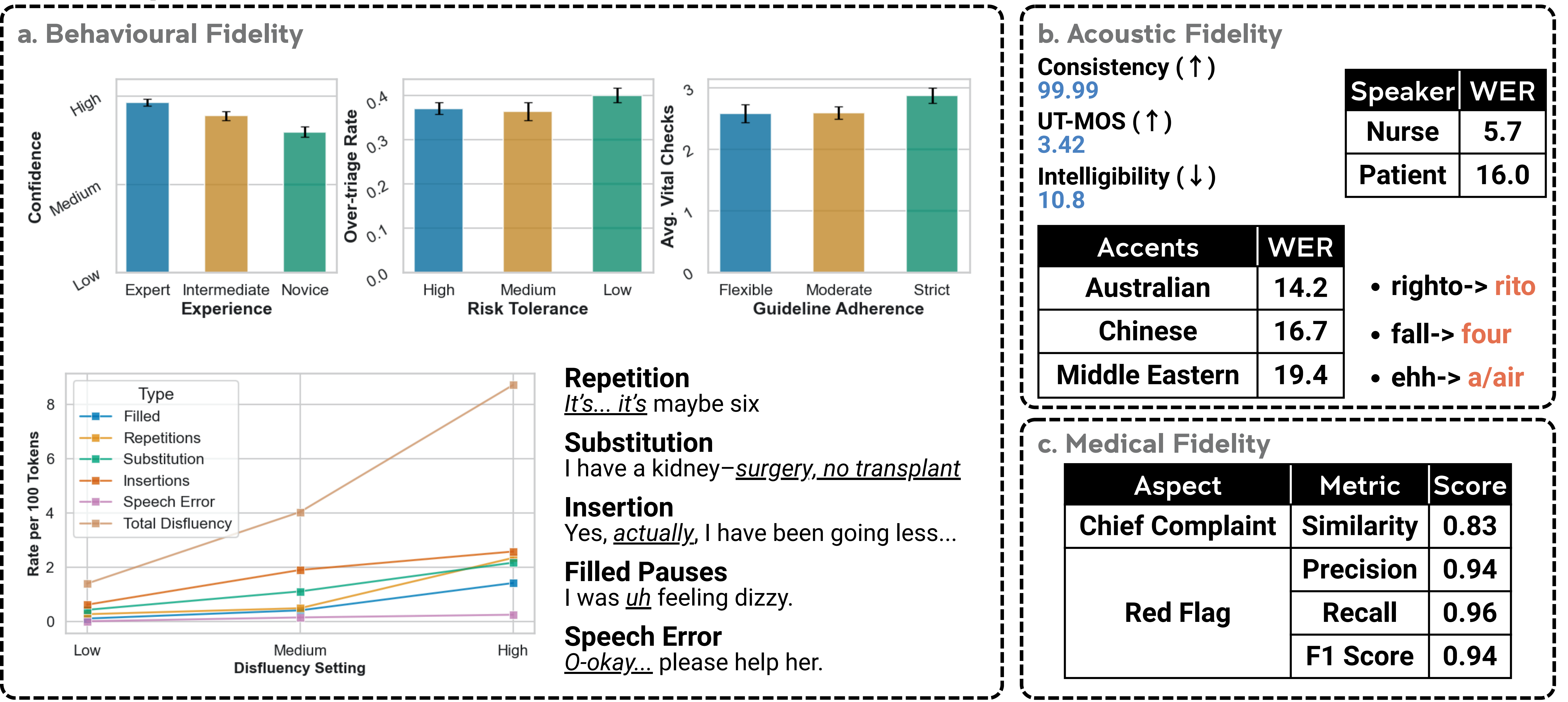
Looking to simulate realistic conversations based on post-hoc notes? Try TriageSim, built for emergency department conversation simulation, by @dipankarsrirag : github.com/dipankarsrirag... #Interspeech2026 #NLPProc

dipankarsrirag/
trriage-sim

1. Framework



2. Quality Evaluation



3. Triage Classification

Model	ATS		ESI	
	SYN	ASR	SYN	ASR
Nemo	0.27	0.24	0.21	0.19
Qwen	0.34	0.32	0.27	0.27
Grok	0.39	0.38	0.30	0.33
Mean	0.33	0.31	0.26	0.25

Model	ATS		ESI	
	SYN	ASR	SYN	ASR
Voxtral	0.28		0.27	

SYN: Generated conversation
ASR: Transcribed conversation

ATS: Australasian Triage Scale
ESI: Emergency Severity Index



arXiv

Package

UNSW-NLP

TRIBOT



Thanks A Bunch!